

Exploring RGB and Video V-JEPA 2.1 Features for Aerial Depth Estimation and Semantic Scene Completion

001 We thank LD5V, CWVJ, and y7qY for their detailed
002 feedback. We agree that unresolved appendix refer-
003 ences and several presentation errors obscured the sub-
004 mitted evidence. We clarify the main result and appro-
005 priately narrow our conclusions below; no unrequested
006 new result is introduced.

007 Presentation and protocol corrections (all reviewers).
008 The appendix tables/figures were not packaged with
009 resolvable references; we apologize because this made
010 supporting evidence inaccessible. Table 3’s caption in-
011 correctly names baselines absent from its body and
012 highlights only RMSE; it should name only the in-
013 cluded DepthAnything2-OccuFly baseline and com-
014 pare every metric. Table 4 is a stress-test result, not a
015 comparative ablation. Equation 19 is correct: training
016 uses 30/50 m, while 40 m validation uses scenes 01–04
017 and testing uses 05–08; “1–9” in Table 4 is an error.
018 Because these scene identities occur at other training
019 altitudes, this holds out altitude, not scene identity.
020 We therefore retract the scene-generalization interpre-
021 tation. We will repair all references, cite every figure,
022 verify figure text size, and replace the screenshot-like
023 overview with an end-to-end data-flow diagram distin-
024 guishing frozen and trainable components.

025 What the depth comparison establishes (CWVJ,
026 y7qY). FiLM-DPT-RGB versus DepthAnything2-
027 OccuFly improves RMSE (4.621 vs. 4.844 m), MAE
028 (3.523 vs. 4.193 m), and AbsRel (0.129 vs. 0.134), is es-
029 sentially tied but slightly worse on δ_1 (0.833 vs. 0.834),
030 and is worse on SILog (0.164 vs. 0.112). RMSE/MAE
031 measure absolute metric error (with RMSE empha-
032 sizing large errors), whereas δ and SILog measure
033 multiplicative/log-space consistency. Thus the sup-
034 ported conclusion is competitive absolute metric accu-
035 racy from a frozen general-purpose backbone, not over-
036 all superiority. The metric pattern is an observation;
037 without error-distribution analysis we do not assert its
038 mechanism.

039 FiLM, backbone, and uncertainty scope (LD5V,
040 CWVJ). We agree that the unexpectedly weak plain-
041 DPT row prevents a clean causal attribution to FiLM.
042 We consequently restrict the claim to the configured
043 FiLM-DPT system being the strongest tested V-JEPA
044 head; Table 4 alone does not isolate conditioning.
045 Likewise, one frozen encoder cannot demonstrate V-
046 JEPA-specific superiority, so conclusions are limited
047 to V-JEPA 2.1 under this OccuFly protocol. Matched

DPT/FiLM conditioning ablations, a second frozen
backbone, and multi-seed uncertainty are the correct
requested controls; we do not present unavailable re-
sults as established evidence.

Video, MVS, and SSC interpretation (LD5V, CWVJ,
y7qY). The strongest submitted SSC result is 37.68
SC IoU / 3.46 SSC mIoU; the strongest Pose-MVS re-
sult is 32.55 / 2.36. Hence video/MVS does not im-
prove the simpler RGB-context configuration, and we
retract any implication of a new superior SSC architec-
ture. DTA does not explicitly warp temporal features
into the target camera frame, so UAV ego-motion can
cause spatial mixing. Pose-MVS was evaluated only
through downstream SSC, not direct depth; it is there-
fore an exploratory negative result, not evidence that
MVS depth itself fails. Direct depth, pose-aligned tem-
poral aggregation, oracle lifting, per-class analysis, and
repeated seeds are needed to localize the bottleneck.
Finally, eight H100s describe experimental infrastruc-
ture, not demonstrated onboard deployment; we will
remove that implication and report efficiency before
making it.

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